

DIRseq: a method for predicting drug-interacting residues of intrinsically disordered proteins from sequences

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eLife Assessment

This **important** study presents a sequence-based method for predicting drug-interacting residues in intrinsically disordered proteins (IDPs), addressing a significant challenge in understanding small-molecule:IDP interactions. The findings have **solid** support through examples underscoring the role of aromatic interactions. While predicted binding sites remain coarse, validation was done on a total of 10 IDPs at varying depths. The method builds on the authors' previous work and, with ad hoc modifications, is poised to benefit this emerging field.

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Abstract

Intrinsically disordered proteins (IDPs) are now well-recognized as drug targets. Identifying drug-interacting residues is valuable for both optimizing compounds and elucidating the mechanism of action. Currently, NMR chemical shift perturbation and all-atom molecular dynamics (MD) simulations are the primary tools for this purpose. Here we present DIRseq, a fast method for predicting drug-interacting residues from the amino-acid sequence. All residues contribute to the propensity of a particular residue to be drug-interacting; the contributing factor of each residue has an amplitude that is determined by its amino-acid type and attenuates with increasing sequence distance from the particular residue. DIRseq predictions match well with drug-interacting residues identified by NMR chemical shift perturbation and other methods, including residues L₂₂WK₂₄ and Q₅₂WFT₅₅ in the tumor suppressor protein p53. These successes augur well for deciphering the sequence code for IDP-drug binding. DIRseq is available as a web server at <https://zhougroupp-uic.github.io/DIRseq/> and has many applications, such as virtual screening against IDPs and designing IDP fragments for in-depth NMR and MD studies.

Introduction

Intrinsically disordered proteins (IDPs) are now recognized as important drug targets ¹⁻⁴. For structured protein targets, a crucial step is identifying the drug-binding pocket. Although an IDP can be locked into a specific conformation by a drug molecule in rare cases ⁵, the prevailing scenario is that the protein remains disordered upon drug binding ⁶⁻¹⁴. Consequently, the IDP-drug complex typically samples a vast conformational space, and the drug molecule only exhibits preferences, rather than exclusiveness, for interacting with subsets of residues ^{8,12,13}. Such drug-interacting residues, akin to binding pockets in structured proteins, are key to optimizing compounds ^{13,15,16} and elucidating the mechanism of action ^{6,9,12,15,16}.

NMR chemical shift perturbation (CSP) is the best experimental method for identifying drug-interacting residues and has been applied to many IDPs ^{7,9,11,26}. Aromatic residues are frequently among drug-interacting residues ^{12,13,15,16,18,20,24}. This is understandable as drug molecules typically are rich in aromatic rings (**Figure 1** and Table S1), which can form π - π interactions with aromatic residues. As recently pointed out by Heller et al. ¹⁴, CSPs of IDPs elicited by drug binding can be small enough to fall within the spectral resolution of NMR spectroscopy, therefore making it difficult to unequivocally identify drug-interacting residues. The small magnitude of CSPs may arise because the drug does not have a strong preference for interacting with any residues. Another scenario may be that drug binding induces a conformational shift such as secondary structure formation or even partial folding ^{9,11,13}, so CSPs spread from the directly interacting residues to the rest of the IDP, adding to the difficulty in identifying drug-interacting residues.

All-atom molecular dynamics (MD) simulations have presented atomic details in many IDP-drug systems ^{8,10,12,13,16,24,27-31}. These simulation studies have highlighted the frequent engagement of drug molecules with aromatic residues, particularly in π - π interactions ^{12,13,16,24,28-31}. MD simulations also revealed the emergence of a compact sub-population of the N-terminal disordered region of the tumor suppressor protein p53 upon binding the drug epigallocatechin-3-gallate (EGCG) ¹² and reduced solvent exposure of hydrophobic residues in the 42-residue amyloid- β peptide (A β 42) upon binding the drug 10074-G5 ¹⁰. However, MD simulations of IDPs still suffer from the perennial issues of inaccurate force fields and insufficient sampling, which are exacerbated when drug molecules are included. For example, two replicate simulations may identify different drug-interacting residues ³².

Here we present a sequence-based method, DIRseq, as a complement to NMR CSP and MD simulations. This method was motivated by our observation that drug-interacting residues seem to overlap with residues exhibiting elevated transverse relaxation rates (R_2) ^{33,34}, as exemplified by the C-terminal 20 residues of α -synuclein ¹³. Elevated R_2 in IDPs is caused by either local inter-residue interactions or residual secondary structure ³⁵. Based on this understanding, we developed a sequence-based method, SeqDYN, to predict R_2 of IDPs ³³. As suggested previously ³⁴, the propensities of residues to form intramolecular interactions and therefore elevate R_2 should be similar to those for forming intermolecular interactions with drug molecules. DIRseq is therefore an adaptation of SeqDYN. DIRseq predictions match well with drug-interacting residues identified by NMR chemical shift perturbation and other methods, including residues L₂₂WK₂₄ and Q₅₂WFT₅₅ in p53 and C-terminal residues in α -synuclein. DIRseq is available as a web server at <https://zhougroup-uic.github.io/DIRseq/> and has many applications.

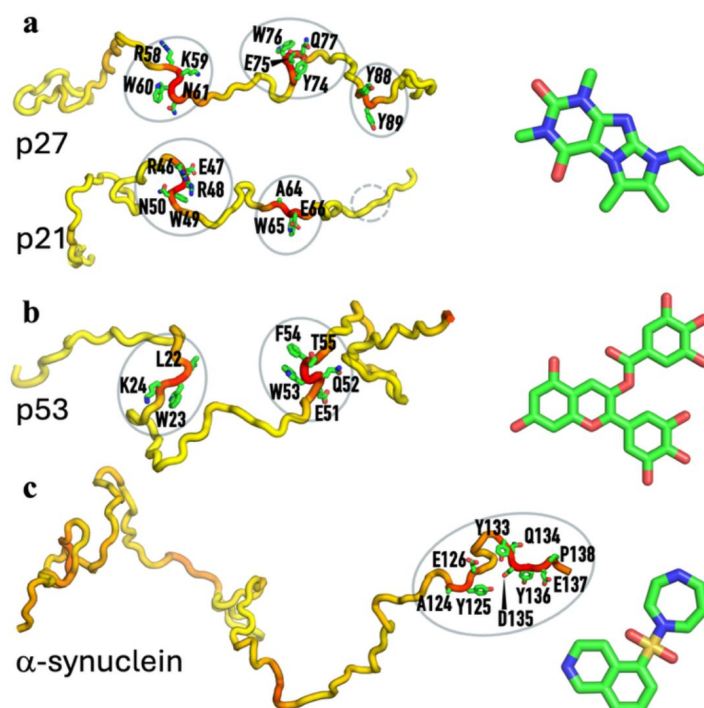


Figure 1.

Four IDPs and drugs that bind to them.

(a) p27, p21, and SJ403. (b) p53 and EGCG. (c) α-synuclein and Fasudil. For each IDP, the DIRseq propensities are rendered by a color spectrum from yellow for low values to red for high values. Predicted drug-interacting residues are shown with sidechains rendered in stick.

Results

Retooling of SeqDYN into DIRseq

SeqDYN ³³ predicts the R_2 value of residue n by accumulating a contributing factor $f(i; n)$ from each residue i :

$$R_2(n) = Y \prod_{i=1}^N f(i; n) \quad (1a)$$

where N is the total number of residues in the IDP and Y is a uniform scale parameter. The contributing factor $f(i; n)$ has an amplitude $q(i)$ that is determined by the amino-acid type of residue i and attenuates with increasing sequence distance, $s = |i - n|$, from residue n :

$$f(i; n) = 1 + \frac{q(i) - 1}{1 + bs^2} \quad (1b)$$

The 20 q parameters (one for each amino-acid type) and b were trained on the measured R_2 values for 45 IDPs. The original q values (Figure S1a) show that aromatic (Trp, Tyr, His, and Phe), Arg, and long aliphatic (Ile and Leu) amino acids are interaction-prone and tend to elevate R_2 . (The original SeqDYN also has an option of applying a helix boost; we did not apply this boost here.) We transformed SeqDYN into DIRseq by implementing four changes.

First, we reassigned four q parameters. We lowered the q values of the three long aliphatic amino acids, Leu, Ile, and Met, to the q value of a short aliphatic amino acid, Val, because long aliphatic amino acids primarily participate in hydrophobic interactions, which may be less important for stabilizing the binding of a small molecule in sites largely exposed to water. At the same time, we increased the q value of Asp to be the same as that of Glu, to increase the role of Asp's electrostatic interactions. Both the downgrade of hydrophobic interactions and the upgrade of electrostatic interactions were motivated by observations on the drug binding to α -synuclein in MD simulations by Robustelli et al. ¹³. The modified set of q parameters is displayed in Figure S1b.

Second, the original b value, 0.0316, corresponds to a correlation half-length, $b^{-1/2}$, of 5.6 residues. Given the small size of drug molecules, we increased the b value to 0.3, corresponding to a correlation half-length of 1.8 residues. That is, we expect a drug molecule to interact with 3 to 4 residues at a time. Third, the SeqDYN-predicted R_2 profile, capturing experimental observations ³⁶, falls off at both termini, because no residues beyond the termini are present to provide a contributing factor to R_2 . However, whereas intra-IDP interactions experience such a terminal effect, IDP-drug interactions do not. To eliminate the terminal effect, we padded the original IDP sequence by a stretch of 12 Gln residues at each terminus. Gln was selected because its q value is at the middle of the 20 q values (Figure S1b).

Lastly, we converted high and low R_2 values into high and low drug-interacting propensities, respectively, using a sigmoid function

$$P = \frac{100}{1 + e^{-\frac{R_2 - R_{2th}}{R_{2wd}}}} \quad (2)$$

where the midpoint R_{2th} and width R_{2wd} of the transition region are determined by the mean (m) and standard deviation (SD) of the R_2 values over the entire sequence. Specifically,

$$R_{2th} = m + s_1 SD \quad (3a)$$

$$R_{2wd} = m/s_2 \quad (4a)$$

We chose s_1 to be 1.5 and s_2 to be 14.0. On the DIRseq web server, users can either keep these default values for b , s_1 , and s_2 , or enter values of their choice. There is also a link for users to download the source code. The values of P range from 0 to 100; $P = 50$ when R_2 is at the “threshold” value R_{2th} . Residues with $P \geq 50$ are predicted to be drug-interacting, but an isolated residue with $P \geq 50$ is discarded, as it can be reasonably expected that interactions with at least two consecutive (or nearby) residues are needed to generate sufficient drug binding stability.

Detailed assessment of DIRseq on four IDPs

Drug-interacting residues in four IDPs or intrinsically disordered regions (IDRs) have been thoroughly characterized by NMR CSP. In **Figure 1** (left images), we show these IDPs in a single conformation, with DIRseq propensities rendered in a color spectrum (from low in yellow to high in red); predicted drug-interacting residues (i.e., those with $P \geq 50$) are shown with sidechains in stick. Drug molecules that bind to these IDPs are also displayed in **Figure 1** (right images). We display NMR CSPs as blue bars and DIRseq propensities as a red curve in **Figure 2**. Unless otherwise noted, we use $m + 1.5$ SD as the threshold for identifying drug-interacting residues, for both CSP and DIRseq; this threshold is indicated by a horizontal dashed line in **Figure 2**. Experimentally identified drug-interacting residues are further indicated by cyan shading. Below we present a detailed comparison between CSP and DIRseq.

The kinase inhibitory domain (residues 22-105) of the cell cycle regulator p27 harbors three aromatic-centered motifs that interact with a group of compounds represented by SJ403 (**Figure 1a**), as found by Iconaru et al.¹⁵ These motifs, $W_{60}N_{61}$, $E_{75}WQ_{77}$, and $Y_{88}Y_{89}$ (**Figure 2a**, cyan shading), are correctly predicted by DIRseq (**Figure 2a**, portions of the red curve above the 50% threshold). Specifically, the predicted drug-interacting residues are $R_{58}KWN_{61}$, $Y_{74}EWQ_{77}$, and $Y_{88}Y_{89}$ (**Figure 1a**, left image). Moreover, the DIRseq propensity profile matches well the CSP profile along the sequence (**Figure 2a**). The two parameters show a very strong correlation, with a Pearson correlation coefficient (r) of 0.82 (Figure S2a).

Iconaru et al. also studied SJ403 binding to a related cell cycle regulator p21. Their CSP data identified the first two corresponding motifs, $W_{49}N_{50}$ and $F_{63}A_{64}$, but not the third as drug-interacting residues (**Figure 2b**, cyan shading). DIRseq correctly predicts both the first two motifs above the 50% threshold and the third motif below the threshold (**Figure 2b**, red curve; **Figure 1b**, left image). As noted by Iconaru et al., the p21 counterpart to the p27 third motif $F_{87}YY_{89}$ is $L_{76}YL_{78}$; the two Leu residues in p21 are no match for F_{87} and Y_{89} in p27. For the DIRseq method, Leu has a lower q value than Tyr and Phe (Figure S1b). CSP and DIRseq propensity show a strong correlation ($r = 0.66$; Figure S2b).

NMR CSP revealed two motifs, $W_{23}K_{24}$ and $D_{49}IEQWFT_{55}$, in the N-terminal region of p53 as interacting with EGCG¹² (**Figure 2c**, cyan shading). MD simulations supported these drug-binding sites. DIRseq again correctly predicts these two motifs (**Figure 2c**, red curve). Specifically, the predicted drug-interacting motifs are $L_{22}WK_{24}$ and $E_{51}QWFT_{55}$ (**Figure 1c**, left image). Here also CSP and DIRseq propensity show a strong correlation ($r = 0.67$; Figure S2c).

Using the $m + 1.5$ SD threshold, CSPs of Robustelli et al.¹³ select residues $A_{124}YEMPS_{129}$ and $Y_{133}QDYE_{137}$ of α -synuclein as interacting with the drug Fasudil (**Figure 2d**, cyan shading). These two C-terminal motifs are also identified by DIRseq, with residues $A_{124}YE_{126}$ and $Y_{133}QDYE_{138}$ above the 50% threshold (**Figure 2d**, red curve; **Figure 1d**, left image). There is a moderate correlation ($r = 0.51$; Figure S2d) between CSP and DIRseq propensity. Robustelli et al. further characterized the IDP-drug interactions by all-atom simulations, using both full-length α -synuclein and a C-terminal fragment (residues 121-140). Further simulations of the C-terminal fragment binding with additional compounds led to a compound known as Ligand 47 with a somewhat higher affinity than Fasudil. The CSP profiles elicited by Fasudil and Ligand 47 are similar, though with larger amplitudes around Y_{125} and Y_{136} by the latter; their correlation coefficient is 0.78 (Figure S3a). Because DIRseq does not consider any information about the drug

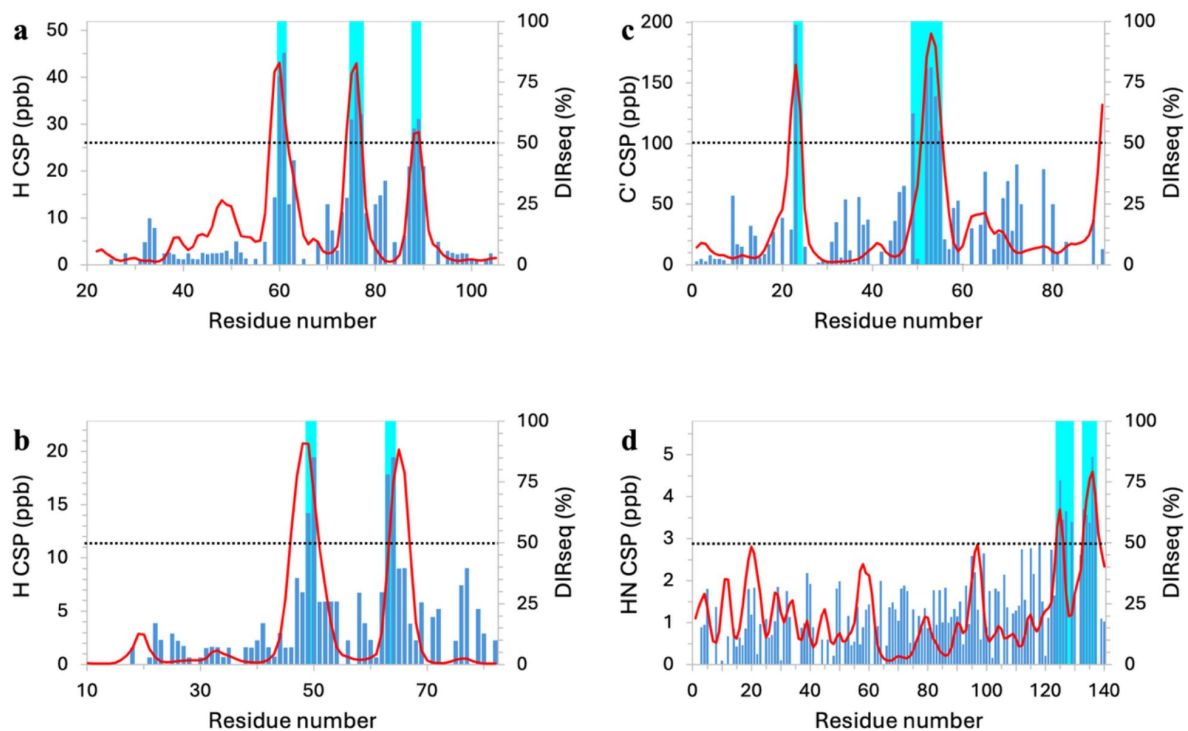


Figure 2.

Comparison of DIRseq propensities with NMR CSPs.

(a) p27. (b) p21. (c) p53. (d) α -synuclein. CSPs are displayed as blue bars and in units of ppb; CSP-identified drug-interacting residues are indicated by cyan shading. DIRseq predictions are shown as red curves. The ordinate scales are chosen so that the $m + 1.5$ SD threshold for CSP is at the same height as the 50% threshold for DIRseq, indicated by a horizontal dashed line.

molecule, this correlation coefficient between CSPs elicited by two different compounds might be viewed as an upper bound of what can be achieved for the correlation between CSP and DIRseq propensity.

Dependences of DIRseq prediction accuracies on model parameters

We now use the CSP data of the above four IDPs to assess the dependences of DIRseq prediction accuracies on model parameters (Table S2). We use two complementary measures for accuracy: the Pearson correlation between CSP and drug-interacting propensity and the difference between true positive (TP) and false positive (FP). The four IDPs have a total of 31 CSP-identified drug-interacting residues. Recall that DIRseq uses a sigmoid function to convert the R_2 output of SeqDYN into drug-interacting propensities. Using the original 20 q parameters along with the present values for b , s_1 , and s_2 , the predictions are already pretty reasonable: the correlations for the four IDPs range from 0.46 to 0.79 (sum = 2.43) and TP outnumbers FP by 7 residues. Indeed, this initial success was anticipated ^{33,34} and validated the premise of DIRseq, i.e., intra-IDP interactions that elevate R_2 also tend to mediate drug interactions. As stated above, we tweaked four of the 20 q parameters to arrive at the final DIRseq; now the r sum increases to 2.66 and TP – FP increases to 15.

Examining the four q parameter changes one at a time, the downgrade of q for Leu increases both r sum (to 2.53) and TP – FP (to 16) relative to the counterparts with the original SeqDYN q parameters. In comparison, the downgrade of q for Ile does not affect r sum and yields a small increase in TP – FP (to 10), the upgrade of q for Asp yields small increases in both r sum (to 2.52) and TP – FP (to 8), but the downgrade of q for Met actually decreases both r sum (to 2.37) and TP – FP (to 6). The latter result shows that we made these parameter tweaks not solely for increasing accuracy but for physical reasons, i.e., to reduce the role of hydrophobic interactions but elevate the role of electrostatic interactions in IDP-drug binding, as suggested by MD simulations.

We also tested alternative models for q parameters. Given the prominence of aromatic amino acids in drug binding as revealed by CSP, we wondered how a model that solely emphasizes aromatic amino acids would perform. To that end, we tested a model with only two different q values for the 20 amino acids: a high value (same as that of Trp in SeqDYN) for all three aromatic amino acids (Trp, Tyr, and Phe), and a low value (same as that of Gly in SeqDYN) for all other amino acids. This is similar to a sticker-spacer model for simulating liquid-liquid phase separation ³⁷. This aromatic model achieves the same r sum, 2.66, as DIRseq, but its FP outnumbers its TP, such that TP – FP = -1. Tesei et al. ³⁸ parameterized a coarse-grained force field to simulate liquid-liquid phase separation. Their λ “stickiness” parameters have a good correlation with the SeqDYN original q parameters ³⁹. We replaced the q parameters by these λ parameters (after scaling to the same numerical range as the q parameters); the resulting model has both a low r sum (1.21) and a low TP – FP (-21). This outcome suggests that, unlike liquid-liquid phase separation where stickiness is the main drive, drug binding is more selective in the type of intermolecular interactions. Lastly, we tested a model where an average hydrophathy scale (compiled by Tesei et al. ⁴⁰) was used in place of the q parameters; the resulting model has very little predictive value (r sum = 0.44 and TP – FP = -20). This last outcome is in line with our downgrade of hydrophobic interactions in DIRseq. The test results from all these alternative models indicate that the q parameters in the final DIRseq model capture the appropriate balance among aromatic, electrostatic, and hydrophobic interactions in IDP-drug binding.

In addition to the 20 q parameters, DIRseq has three other parameters. The b parameter determines the number of residues that simultaneously interact with a drug molecule. In Table S2, we list the performance measures when b is varied. These results show that our final choice, 0.3, for b is optimal for both r sum and TP – FP. The corresponding number of residues, 3 to 4, that simultaneously interact with a drug molecule fits with the narrow range of drug molecule sizes in

the present study (Table S2; molecular weights: 360 ± 130 Da). The two last parameters, s_1 and s_2 , are in the sigmoid function that converts the R_2 output into drug-interacting propensities. s_1 sets the R_2 threshold for labeling a residue as drug-interacting. Again, our final choice, corresponding to $m + 1.5$ SD for the threshold, achieves an optimum for r sum and TP – FP. A lower threshold leads to a high FP and also a slight deterioration in r sum, whereas a higher threshold leads to a low FP and possibly a minuscule improvement in r sum. s_2 controls the sharpness of the sigmoid function in the transition region, and affects r sum but not TP – FP. r sum increases with increasing s_2 ; we chose an s_2 value where r sum is nearly at the saturation level.

DIRseq as a complement to CSP for assigning drug-interacting residues

After assessing the achievable accuracy of DIRseq, we now consider an application: its combination with CSP to make robust assignments of drug-interacting residues. We note that the above CSP-identified drug-interacting motifs are all anchored on one or more aromatic residues, and this feature likely contributes to the good performance of DIRseq. When a clear “aromatic signal” is not present, CSP or DIRseq alone may not be able to conclusively identify drug-interacting residues. However, a consensus identification by CSP and DIRseq may be reliable. MD simulations have played such a complementary role to CSP in several studies ^{12,13,16,30}.

De Mol et al. ⁹ reported the CSPs of an IDR called AF-1* (residues 142–448) in the androgen receptor elicited by EPI-001 and its stereoisomers including EPI-002; small but reproducible CSPs were found in three subregions, R1 (residues 341–371), R2 (residues 391–414), and R3 (residues 426–446). Conversely, AF-1* caused changes in the ¹H NMR spectrum of EPI-001 but the peptides corresponding to R1–R3 did not, suggesting that, rather than separately interacting with the three subregions, EPI-001 induces a partial folding of the three subregions. MD simulations captured the partial folding of the R2–R3 fragment (residues 391–446) induced by EPI-002, and underscored the importance of aromatic residues in drug binding ³⁰. Basu et al. ¹⁶ reported the CSPs of the transactivation unit 5 (Tau-5*, residues 336–448) in AF-1* by EPI-001 and a more potent variant, 1aa (**Figure 3a** and **Figure S3b**). The contact probabilities of the R2–R3 fragment with 1aa in MD simulations reaffirmed the importance of aromatic residues.

De Mol et al. ⁹ and Basu et al. ¹⁶ were careful not to name any drug-interacting residues based on CSPs. In **Figure 3a**, we compare the 1aa-elicited CSPs and DIRseq propensities of Tau-5*. In agreement with both NMR studies, DIRseq identifies drug-interacting residues in the middle of each of R1–R3: R₃₆₀DYY₃₆₃, A₃₉₆WAA₃₉₉, and W₄₃₃H₄₃₄. In addition, the latter two motifs showed the highest drug-contact probabilities in separate MD simulations of the R2–R3 fragment ^{16,30}. For the R2 and R3 subregions, CSPs above the $m + 1.5$ SD threshold are observed at residues downstream of the DIRseq identifications, so we propose to expand the drug-interacting motifs to A₃₉₆WAAAAAQ₄₀₃ and W₄₃₃HTLF₄₃₇. The three putative drug-interacting motifs are indicated as cyan shading in **Figure 3a**.

Heller et al. ¹⁴ used ¹⁹F transverse relaxation measurements to determine the binding affinity of the disordered domains 2 and 3 of the hepatitis C virus NS5A protein (NS5A-D2D3, residues 247–466) for 5-fluorindole. They also measured ¹H-¹⁵N CSPs at two ligand concentrations but described them as “nearly undetectable” (**Figure 3b**). We speculate that the small CSPs may be due to the small size of the ligand, making it difficult to interact with multiple residues simultaneously and thus achieve sufficient binding stability. In any event, CSPs above the $m + 1.5$ SD threshold were largely isolated (and thus appeared to be random), and there was not much overlap between residues having these above-the-threshold CSPs at the two ligand concentrations. The one exception is a motif around W₃₁₂, which had above-the-threshold CSPs at both ligand concentrations, and nearby residues A₃₀₈ and A₃₁₃ also had above-the-threshold CSPs at one of the ligand concentrations. DIRseq predicts the motif P₃₁₀AWARPD₃₁₆ as drug-interacting residues. We

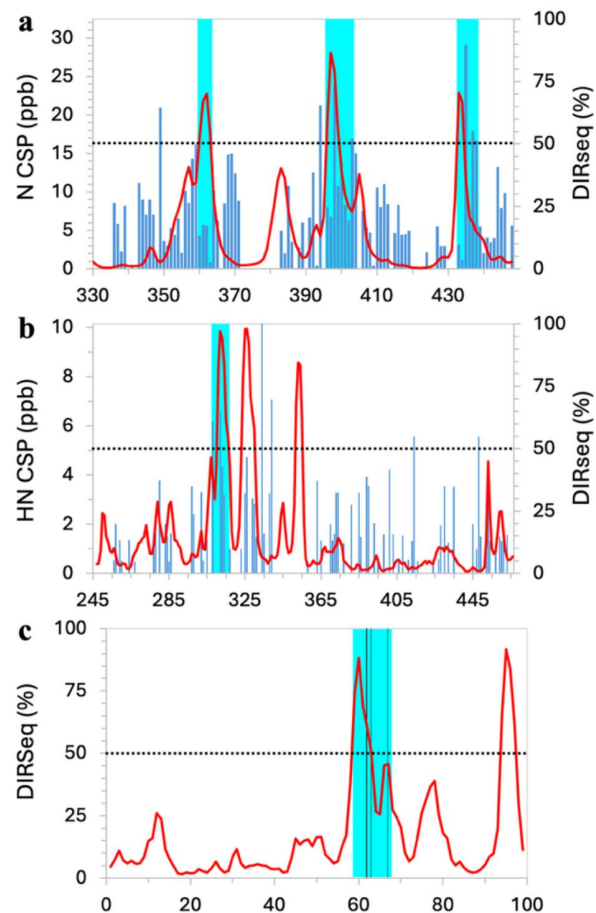


Figure 3.

Drug-binding sites identified by combining CSP or mutation data with DIRseq predictions.

(a) Tau-5*. (b) NS5A-D2D3. (c) β2 microglobulin. Display items in panels (a, b) have the same meanings as in [Figure 2](#), except that cyan shading indicates consensus identification; in panel (c), vertical lines indicate mutation sites.

propose the expanded motif, A₃₀₈LP_{AWARPD}₃₁₆, as residues interacting with 5-fluorindole (**Figure 3b** [↗](#), cyan shading). DIRseq also predicts two more motifs, E₃₂₃SWRRPDY₃₃₀ and R₃₅₂RRR₃₅₅, as drug-interacting residues, which remain to be tested.

The fibrillation of acid-denatured β_2 microglobulin is inhibited by rifamycin SV [↗](#). An aromatic-rich motif, W₆₀SFYLLYYTEF₇₀, was implicated in the nucleation of fibrillation and also involved in ligand binding, as a triple mutation, F62A/Y63A/Y67A, significantly weakened binding. Low intensities of NMR peaks from residues 58-79 (possibly due to the formation of residual structures) prevented the measurement of CSPs. DIRseq predicts D₅₉WSFY₆₃ as drug-interacting residues. Combined with the mutational data of Woods et al. [↗](#), we propose the expanded motif, D₅₉WSFYLLYY₆₇, as the major drug-binding site (**Figure 3c** [↗](#), cyan shading). DIRseq also predicts an additional motif, K₉₄WDR₉₇, as drug-interacting residues.

The aggregation of human islet amyloid polypeptide (hIAPP; 37 residues) is inhibited by the small molecule YX-I-1 [↗](#). CSPs elicited by this molecule were small (**Figure 4a** [↗](#)). In addition, CSPs of short IDPs may not exhibit strong disparities because amino acids may be too well mixed along the sequence or drug binding may induce a conformational shift. We thus reduce the threshold for identifying drug-interacting residues to $m + 1.0$ SD when the number of residues is ≤ 50 . With this threshold, three residues are identified by CSP as drug-interacting residues: R₁₁ and V₁₇H₁₈. In comparison, DIRseq identifies T₉QRLA₁₃ and F₁₅ as drug-interacting residues, which partially overlap with the CSP identifications. Combining the two types of data, we propose T₉QRLANFLVH₁₈ as the primary drug-binding site (**Figure 4a** [↗](#), cyan shading). We note that this motif is also prone to α -helix formation [↗](#).

Ono et al. [↗](#) acquired [↗](#)H-[↗](#)N heteronuclear single quantum coherence spectra of the 42-residue amyloid- β (A β 42) in the absence and presence of the oligomerization-blocking compound myricetin. CSPs were not calculated, but chemical shift movements were most pronounced at R₅, V₁₂H₁₃, K₁₆LVFFAED₂₃, and I₃₁I₃₂ (vertical lines in **Figure 4b** [↗](#)). In addition, NMR cross peaks suffered broadening upon ligand binding at four of these residues: R₅, V₁₂, K₁₆, and V₁₈ (dark vertical lines in **Figure 4b** [↗](#)), implicating elevated probabilities of ligand interactions. DIRseq predicts E₃FRH₆ and H₁₃H₁₄ as drug-interacting residues. Combined with the NMR data, we propose E₃FRH₆ and V₁₂HHQKL₁₈ as the primary ligand-binding sites (**Figure 4b** [↗](#), cyan shading). A β 42 CSPs elicited by several other compounds were also widely distributed over the sequence, such that Iwaya et al. [↗](#) “failed to identify” drug-interacting residues, implicating a conformational shift.

Hammoudeh et al. [↗](#) identified three distinct drug binding sites in a C-terminal IDR (residues 363-412) of the oncoprotein c-Myc. This region is disordered on its own but forms a helix-loop-helix structure upon heterodimerization with Max. These authors measured the CSPs of three overlapping fragments (residues 363-381, 370-409, and 402-412) each in the presence of a single compound and of the full IDR in the presence of all three compounds. Using 20 parts per billion (ppb) as the threshold, Hammoudeh et al. named residues R₃₆₆RNELKRSFF₃₇₅, F₃₇₅ALRDQIPELE₃₈₅, and Y₄₀₂ILSVQAE₄₀₉ as the binding sites for 10074-G5, 10074-A4, and 10058-F4, respectively. We present their CSP data for the full IDR in **Figure 4c** [↗](#). Using the $m + 1.0$ SD threshold, only six residues are identified as drug-interacting residues: R₃₆₇, L₃₇₀, F₃₇₄F₃₇₅, A₃₉₉, and S₄₀₅. DIRseq predicts two motifs, E₃₆₃RQRR₃₆₇ and K₃₇₁RSFFA₃₇₆, that overlap with the first two sites identified by CSP; a third motif, K₃₉₇KATAY₄₀₂, that corresponds to the third CSP-identified site has moderate drug-interacting propensities. Combining CSP and DIRseq, we revise the three drug-binding sites to be E₃₆₃RQRRNELKRSFF₃₇₅, S₃₇₃FFALRDQI₃₈₁, and K₃₉₇KATAYILS₄₀₅ (**Figure 4c** [↗](#), cyan, olive, and yellow shading, respectively). In addition to 10074-G5, 10074-A4, and 10058-F4, many compounds bind to these three sites [↗](#), [↗](#), [↗](#).

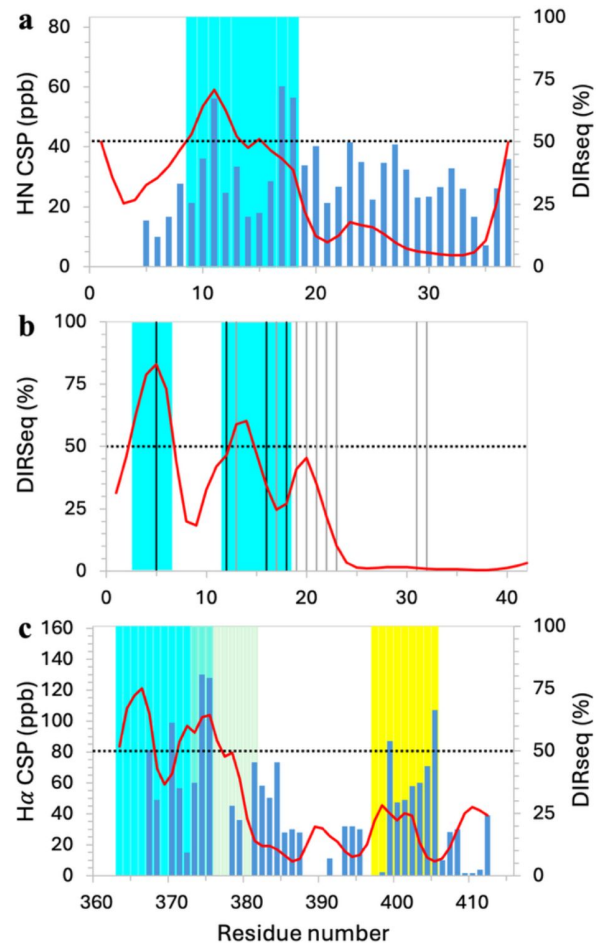


Figure 4.

Drug-binding sites identified by combining CSP or mutation data with DIRseq predictions.

(a) hIAPP. (b) Aβ42. (c) c-Myc. Display items have the same meanings as in [Figure 2](#), but with the following exceptions. (1) In panel (b), vertical lines indicate residues with prominent CSPs; those accompanied by NMR peak broadening have their vertical lines in dark color. (2) In panel (c), three CSP-identified drug-interacting regions are indicated by cyan, olive, and yellow shading. (3) The threshold for identifying drug-interacting residues is lowered to $m + 1.0$ SD.

Discussion

We have presented the first sequence-based method, DIRseq, to predict drug-interacting residues of IDPs. Assessment against NMR CSP demonstrates the accuracy of DIRseq. Drug-binding motifs are anchored on one or more aromatic residues, for forming π - π interactions with drug-like molecules that are rich in aromatic groups. The success of DIRseq comes without any specific information on the drug molecules, suggesting that IDPs may have a relatively simple sequence code for drug binding.

The notion that drug-interacting residues may be agnostic to the molecular details of drug compounds is supported by the fact that the same drug can bind to different IDPs. For example, in addition to p53¹² (Figure 2c), the polyphenol EGCG also binds to many other IDPs, including α -synuclein⁴⁵, hIAPP^{46,47}, A β 40²¹ and A β 42⁴⁵, tau⁴⁸, and merozoite surface protein 2⁴⁹. Likewise, 10074-G5 binds to c-Myc¹⁷ (Figure 4c) but also to A β 42¹⁰. On the other hand, c-Myc represents a case where different compounds bind to distinct sites on a single IDP¹⁷. A related example is presented by p27, where SJ403 typifies a group of compounds that share the same three binding sites (Figure 2a). Another group of compounds, typified by SJ710, binds only to the third site. Chemically, the presence of nitrogen atoms in the rings of SJ403 enhances its aromaticity and thus strengthens π - π interactions; in addition, the electronegative groups of SJ403 project into different directions, making it less restricted when forming electrostatic interactions (Figure S4). These features may explain why SJ403 can bind to all three sites whereas SJ710 can bind only to the third site, F₈₇YYR₉₀, where three consecutive aromatic residues followed by a basic residue ensure that SJ710 can form both π - π and electrostatic interactions. When more data for multiple drugs binding to a single IDP become available, it will be important to use such data to train the next generation of DIRseq where the parameters are drug-specific. As a simple example, the number of residues that can simultaneously bind a drug molecule may grow with the latter's size; this dependence can be modeled by making the parameter *b* dependent on drug molecule size. The drug molecules studied in the present work have molecular weights of 360 ± 130 Da and thus span a relatively narrow size range.

We have illustrated the combination of DIRseq with NMR CSP to make robust identifications of drug-binding sites in IDPs. Indeed, CSP, MD simulations, and DIRseq are three orthogonal approaches that have great potential in complementing each other, not only for identifying drug-binding sites but also for elucidating the roles of amino acids, their sequence context, and different types of noncovalent interactions in forming such sites. DIRseq offers fast speed and a simple, direct link between sequence motif and propensity for drug binding. As more data from CSP and MD simulations become available, DIRseq can be tuned to make even better predictions. For example, it may be feasible to develop DIRseq versions specific to different pharmacophores.

Another application of DIRseq is to define IDP fragments for in-depth study by MD simulations, as shorter constructs both enable the use of a smaller simulation box and reduce the size of the conformational space. For example, based on CSP data and initial MD simulations of full-length of α -synuclein, Robustelli et al.¹³ chose a 20-residue C-terminal fragment for simulations of binding with additional compounds, leading to the identification of Ligand 47 as a stronger binder than the original Fasudil. Similarly, based on CSP data from longer constructs of the androgen receptor, both Zhu et al.³⁰ and Basu et al.¹⁶ chose the 56-residue R2-R3 fragment for MD simulations of drug binding. DIRseq can now play a similar role in selecting fragments for MD simulations when CSP data are unavailable. Longer constructs may also present challenges such as resonance assignments to NMR experiments, so well-chosen fragments guided by DIRseq can also benefit NMR studies.

Lastly, virtual screening has been conducted against conformational ensembles of IDPs [24](#), [50](#); drug-binding sites predicted by DIRseq can be used to guide such screening. As a simple illustration, we present poses of EGCG generated by screening against the two DIRseq-predicted binding sites in p53 in **Figure 5**. As IDPs sample a vast conformational space, knowledge of the binding site can drastically reduce the computational cost. The subset of conformations that generate high docking scores for a given drug at the known site can also provide insight into the mechanism of drug action.

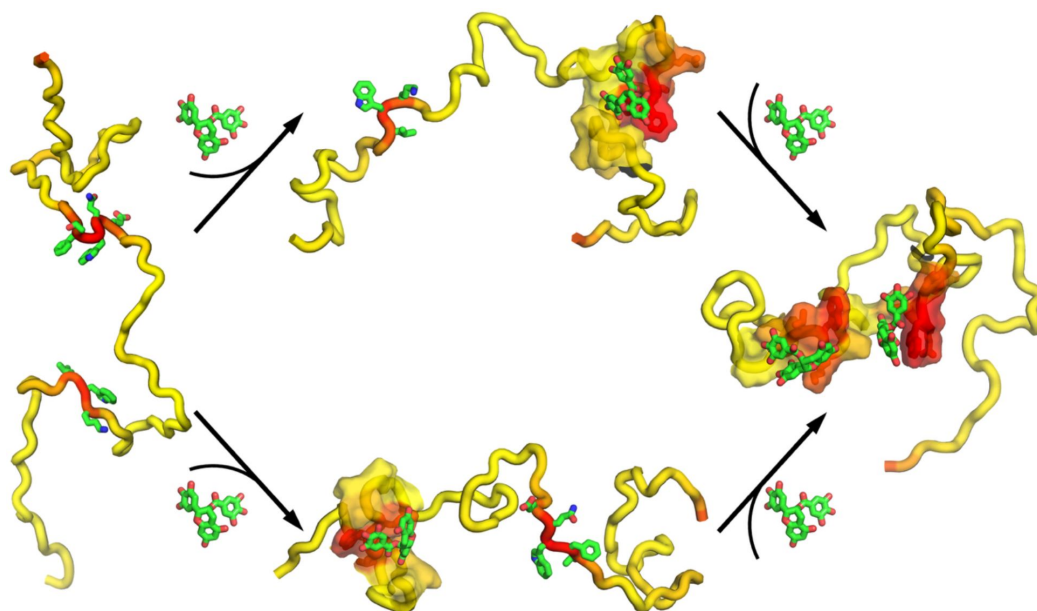


Figure 5.

Poses of p53-bound EGCG generated by docking.

Computational Methods

The sequences of the IDP studied here and the drugs that bind to them are listed in Table S1. All DIRseq predictions were obtained using the web server at <https://zhougroup-uic.github.io/DIRseq/>. Conformations of IDPs were generated using the TraDES method [51](#).

Docking of EGCG onto p53 was performed via the SwissDock web server at <https://www.swissdock.ch/> [52](#), utilizing the Autodock Vina docking engine [53](#). The SMILES string for EGCG from PubChem (<https://pubchem.ncbi.nlm.nih.gov/>; CID 65064) and several conformations of p53 were used as input. A cubic region (13 to 20 Å in side length) around the center of each drug-interacting residue was selected for docking.

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Additional files

Supplementary tables and figures [↗](#)

Additional information

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Reviewer #1 (Public review):

Summary:

The authors developed a sequence-based method to predict drug-interacting residues in IDP, based on their recent work, to predict the transverse relaxation rates (R_2) of IDP trained on 45 IDP sequences and their corresponding R_2 values. The discovery is that the IDPs interact with drugs mostly using aromatic residues that are easy to understand, as most drugs contain aromatic rings. They validated the method using several case studies, and the predictions are in accordance with chemical shift perturbations and MD simulations. The location of the predicted residues serves as a starting point for ligand optimization.

Strengths:

This work provides the first sequence-based prediction method to identify potential drug-interacting residues in IDP. The validity of the method is supported by case studies. It is easy to use, and no time-consuming MD simulations and NMR studies are needed.

Weaknesses:

The method does not depend on the information of binding compounds, which may give general features of IDP-drug binding. However, due to the size and chemical structures of the compounds (for example, how many aromatic rings), the number of interacting residues

varies, which is not considered in this work. Lacking specific information may restrict its application in compound optimization, aiming to derive specific and potent binding compounds.

Comments on revised version:

I'm satisfied with the authors' response and the public review does not need further changes.

<https://doi.org/10.7554/eLife.107470.2.sa2>

Reviewer #2 (Public review):

Summary:

In this work, the authors introduce DIRseq, a fast, sequence-based method that predicts drug-interacting residues (DIRs) in IDPs without requiring structural or drug information. DIRseq builds on the authors' prior work looking at NMR relaxation rates, and presumes that those residues that show enhanced R2 values are the residues that will interact with drugs, allowing these residues to be nominated from the sequence directly. By making small modifications to their prior tool, DIRseq enables the prediction of residues seen to interact with small molecules in vivo.

Strengths:

The preprint is well written and easy to follow.

<https://doi.org/10.7554/eLife.107470.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

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The authors developed a sequence-based method to predict drug-interacting residues in IDP, based on their recent work, to predict the transverse relaxation rates (R2) of IDP trained on 45 IDP sequences and their corresponding R2 values. The discovery is that the IDPs interact with drugs mostly using aromatic residues that are easy to understand, as most drugs contain aromatic rings. They validated the method using several case studies, and the predictions are in accordance with chemical shift perturbations and MD simulations. The location of the predicted residues serves as a starting point for ligand optimization.

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This work provides the first sequence-based prediction method to identify potential druginteracting residues in IDP. The validity of the method is supported by case studies. It is easy to use, and no time-consuming MD simulations and NMR studies are needed.

Weaknesses:

The method does not depend on the information of binding compounds, which may give general features of IDP-drug binding. However, due to the size and chemical structures of the compounds (for example, how many aromatic rings), the number of interacting

residues varies, which is not considered in this work. Lacking specific information may restrict its application in compound optimization, aiming to derive specific and potent binding compounds.

We fully recognize that different compounds may have different interaction propensity profiles along the IDP sequence. In future studies, we will investigate compound-specific parameter values. The limiting factor is training data, but such data are beginning to be available.

Reviewer #2 (Public review):

Summary:

In this work, the authors introduce DIRseq, a fast, sequence-based method that predicts drug-interacting residues (DIRs) in IDPs without requiring structural or drug information. DIRseq builds on the authors' prior work looking at NMR relaxation rates, and presumes that those residues that show enhanced R2 values are the residues that will interact with drugs, allowing these residues to be nominated from the sequence directly. By making small modifications to their prior tool, DIRseq enables the prediction of residues seen to interact with small molecules in vivo.

Strengths:

The preprint is well written and easy to follow

Weaknesses:

(1) The DIRseq method is based on SeqDYN, which itself is a simple (which I do not mean as a negative - simple is good!) statistical predictor for R2 relaxation rates. The challenge here is that R2 rates cover a range of timescales, so the physical intuition as to what exactly elevated R2 values mean is not necessarily consistent with "drug interacting". Presumably, the authors are not using the helix boost component of SeqDYN here (it would be good to explicitly state this). This is not necessarily a weakness, but I think it would behoove the authors to compare a few alternative models before settling on the DIRseq method, given the somewhat ad hoc modifications to SeqDYN to get DIRseq.

Actually, the factors that elevate R2 are well-established. These are local interactions and residual secondary structures (if any). The basic assumption of our method is that intra-IDP interactions that elevate R2 convert to IDP-drug interactions. This assumption was supported by our initial observation that the drug interaction propensity profiles predicted using the original SeqDYN parameters already showed good agreement with CSP profiles. We only made relatively small adjustments to the parameters to improve the agreement. Indeed we did not apply the helix boost portion of SeqDYN to DIRseq, and now state as such (p. 4, second last paragraph). We now also compare DIRseq with several alternative models, as summarized in new Table S2.

Specifically, the authors previously showed good correlation between the stickiness parameter of Tesei et al and the inferred "q" parameter for SeqDYN; as such, I am left wondering if comparable accuracy would be obtained simply by taking the stickiness parameters directly and using these to predict "drug interacting residues", at which point I'd argue we're not really predicting "drug interacting residues" as much as we're predicting "sticky" residues, using the stickiness parameters. It would, I think, be worth the authors comparing the predictive power obtained from DIRseq with the predictive power obtained by using the lambda coefficients from Tesei et al in the model, local density of aromatic residues, local hydrophobicity (note that Tesei et al have tabulated a large set of hydrophobicity scores!) and the raw SeqDYN predictions. In the absence of

lots of data to compare against, this is another way to convince readers that DIRseq offers reasonable predictive power.

We now compare predictions of these various parameter sets, and report the results in Table S2. In short, among all the tested parameter sets, DIRseq has the best performance as measured by (1) strong correlations between prediction scores and CSPs and (2) high true positives and low false positives (p. 7-9).

(2) Second, the DIRseq is essentially SeqDYN with some changes to it, but those changes appear somewhat ad hoc. I recognize that there is very limited data, but the tweaking of parameters based on physical intuition feels a bit stochastic in developing a method; presumably (while not explicitly spelt out) those tweaks were chosen to give better agreement with the very limited experimental data (otherwise why make the changes?), which does raise the question of if the DIRseq implementation of SeqDYN is rather over-parameterized to the (very limited) data available now? I want to be clear, the authors should not be critiqued for attempting to develop a model despite a paucity of data, and I'm not necessarily saying this is a problem, but I think it would be really important for the authors to acknowledge to the reader the fact that with such limited data it's possible the model is over-fit to specific sequences studied previously, and generalization will be seen as more data are collected.

We have explained the rationale for the parameter tweaks, which were limited to q values for four amino-acid types, i.e., to deemphasize hydrophobic interactions and slightly enhance electrostatic interactions (p. 4-5). We now add that these tweaks were motivated by observations from MD simulations of drug interactions with a-syn (ref 13). As already noted in the response to the preceding comment, we now also present results for the original parameter values as well as for when the four q values are changed one at a time.

(3) Third, perhaps my biggest concern here is that - implicit in the author's assumptions - is that all "drugs" interact with IDPs in the same way and all drugs are "small" (motivating the change in correlation length). Prescribing a specific length scale and chemistry to all drugs seems broadly inconsistent with a world in which we presume drugs offer some degree of specificity. While it is perhaps not unexpected that aromatic-rich small molecules tend to interact with aromatic residues, the logical conclusion from this work, if one assumes DIRseq has utility, is that all IDRs bind drugs with similar chemical biases. This, at the very least, deserves some discussion.

The reviewer raises a very important point. In Discussion, we now add that it is important to further develop DIRseq to include drug-specific parameters when data for training become available (p. 12-13). To illustrate this point, we use drug size as a simple example, which can be modeled by making the b parameter dependent on drug molecule size.

(4) Fourth, the authors make some general claims in the introduction regarding the state of the art, which appear to lack sufficient data to be made. I don't necessarily disagree with the author's points, but I'm not sure the claims (as stated) can be made absent strong data to support them. For example, the authors state: "Although an IDP can be locked into a specific conformation by a drug molecule in rare cases, the prevailing scenario is that the protein remains disordered upon drug binding." But is this true? The authors should provide evidence to support this assertion, both examples in which this happens, and evidence to support the idea that it's the "prevailing view" and specific examples where these types of interactions have been biophysically characterized.

We now cite nine studies showing that IDPs remain disordered upon drug binding.

"Consequently, the IDP-drug complex typically samples a vast conformational space, and the drug molecule only exhibits preferences, rather than exclusiveness, for interacting with subsets of residues." But again, where is the data to support this assertion? I don't necessarily disagree, but we need specific empirical studies to justify declarative claims like this; otherwise, we propagate lore into the scientific literature. The use of "typically" here is a strong claim, implying most IDP complexes behave in a certain way, yet how can the authors make such a claim?

Here again we add citations to support the statement.

Finally, they continue to claim:

"Such drug interacting residues (DIRs), akin to binding pockets in structured proteins, are key to optimizing compounds and elucidating the mechanism of action." But again, is this a fact or a hypothesis? If the latter, it must be stated as such; if the former, we need data and evidence to support the claim.

We add citations to both compound optimization and mechanism of action.

Reviewer #1 (Recommendations for the authors):

(1) The authors should compare the sequences of the IDPs in the case studies with the 45 IDPs in training the SeqDYN model to make sure that they are not included in the training dataset or are highly homologous.

Please note that the data used for training SeqDYN were R2 rates, which are independent of the property being studied here, i.e., drug interacting residues. Therefore whether the IDPs studied here were in the training set for SeqDYN is immaterial.

(2) The authors manually tuned four parameters in SeqDYN to develop the model for predicting drug-interacting residues without giving strict testing or explanations. More explanations, testing of more values, and ablation testing should be given.

As responded above, we now both expand the explanation and present more test results.

(3) The authors changed the q values of L , I , and M to the value of V . What are the results if these values are not changed?

These results are shown in Table S2 (entry named SeqDYN_orig).

(4) Only one b value is chosen based on the assumption that a drug molecule interacts with 3-4 residues at a time. However, the number of interacting residues is related to the size of the drug molecule. Adjusting the b value with the size of the ligand may provide improvement. It is better to test the influence of adjusting b values. At least, this should be discussed.

Good point! We now state that b potentially can be adjusted according to ligand size (p. 12-13). In addition, we also show the effect of varying b on the prediction results (Table S2; p. 8, last paragraph).

(5) The authors add 12 Q to eliminate end effects. However, explanations on why 12 Q s are chosen should be given. How about other numbers of Q or using other residues (e.g., the commonly used residues in making links, like GS/PS or A)?

As we already explained, “Gln was selected because its q value is at the middle of the 20 q values.” (p. 5, second paragraph). Also, 12 Qs are sufficient to remove any end effects; a higher number of Qs does not make any difference.

Reviewer #2 (Recommendations for the authors):

(1) The authors make reference to the "C-terminal IDR" in cMyc, but the region they note is found in the bHLH DNA binding domain (which falls from residue ~370-420).

We now clarify that this region is disordered on its own but form a helix-loop-loop structure upon heterodimerization with Max (p. 11, last paragraph).

(2) Given the fact that X-seq names are typically associated with sequencing-based methods, it's perhaps confusing to name this method DIRseq?

We appreciate the reviewer’s point, but by now the preprint posted in bioRxiv is in wide circulation, and the DIRseq web server has been up for several months, so changing its name would cause a great deal of confusion.

(3) I'd encourage the authors just to spell out "drug interacting residues" and retain an IDR acronym for IDRs. Acronyms rarely make writing clearer, and asking folks to constantly flip between IDR and DIR is asking a lot of an audience (in this reviewer's opinion, anyway).

The reviewer makes a good point; we now spell out “drug-interacting residues”.

(4) The assumption here is that CSPs result from direct drug:IDR interactions. However, CSPs result from a change in the residue chemical environment, which could in principle be an indirect effect (e.g., in the unbound state, residues A and B interact; in the bound state, residue A is now free, such that it experiences a CSP despite not engaging directly). While I recognize such assumptions are commonly made, it behoves the authors to explicitly make this point so the reader understands the relationship between CSPs and binding.

We did add caveats of CSP in Introduction (p. 3, second paragraph).

(5) On the figures, please label which protein is which figure, as well as provide a legend for the annotations on the figures (red line, blue bar, cyan region, etc.)

We now label protein names in Fig. 1. For annotation of display items, it is also made in the Figs. 2 and 3 captions; we now add it to the Fig. 4 caption.

(6) abstract: "These successes augur well for deciphering the sequence code for IDP-drug binding." - This is not grammatically correct, even if augur were changed to agree. Suggest rewriting.

“Augur well” means to be a good sign (for something). We use this phrase here in this meaning.

(6) page 5: "we raised the q value of Asp to be the same as that of Glu" → suggested "increased" instead of raised.

We have made the suggested change.

(7) The authors should consider releasing the source code (it is available via the .js implementation on the server, but this is not very transferable/shareable, so I'd encourage the authors to provide a stand-alone implementation that's explicitly shareable).

We have now added a link for the user to download the source code.

<https://doi.org/10.7554/eLife.107470.2.sa0>